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Infrastructure-Driven Development: The Local Social Impact of a Large Hydropower Plant in the Amazon

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ABSTRACT *The Brazilian Amazon is marked by attempts at infrastructure-driven development. The construction of the Belo Monte dam, the third-largest in the world, brought chaotic and rapid urbanization to surrounding cities. This paper answered whether the Belo Monte dam impacted the level of violent crime in the region after Altamira was ranked as the most violent city in Brazil in 2015. Following a difference-in-difference approach, I explore the timing of the Belo Monte dam construction and the distance from the construction site to identify the causal effect of unplanned urbanization on homicide rate. In two exogenous shocks, the beginning (2011) and the end of the construction (2015), I estimated a significant rise in the homicide rate in closer cities. The results are driven by criminal activity, with drug trafficking being one of the channels behind the rising homicide rate during construction. The homicide victims are mainly the young male population causing a significant loss of human capital. The increasing homicide rate after the end of the construction indicates that the Belo Monte dam may have a long-term effect on the violence level in the region. Violence imposes high social costs and may jeopardize future growth in the Amazon.*

KEYWORDS: Economic development; infrastructure-driven development; social impact; violence; hydropower; Amazon

1. Introduction

Large-scale infrastructure projects are meant to achieve progress and growth. Dams can improve agriculture production through irrigation and reduce poverty by causing development (Duflo & Pande, 2007). Additionally, constructing hydropower plants can potentially increase economic activity and tax revenue in the surrounding cities (de Faria, Davis, Severnini, & Jaramillo, 2017). However, the debate around the construction of dams lacks systematic empirical evidence on how these projects distribute their costs and benefits.

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Going in the same direction as Duflo and Pande (2007) and de Faria et al. (2017), this paper aims to provide evidence on how a large dam affects the welfare of the local population. I focus on Brazil, which has approximately 70 percent of its energy hydropower generated (ANEEL, 2019). With 1,361 hydropower plants in operation, 32 in construction, and 111 in the planning stage, Brazil is among the top three countries producing the most hydroelectric power. While hydropower can provide clean energy demanded by a growing nation, each dam also represents socioeconomic impacts on communities around the reservoirs. It remains widely questioned whether dams' distributional and productivity implications are fairly distributed across the population.

Many social impacts caused by hydroelectric dams are reported in the literature, with few studies examining local impacts in developing countries (de Faria et al., 2017; Fearnside, 2001, 2014; Jackson & Sleight, 2000; Sovacool & Bulan, 2011). The economic activity created by a large dam construction is sudden and attracts many people. Between workers and displaced people, cities grow and experience rapid urbanization. When the cities in question are in developing countries, urban size, and weak governments can cause institutional failure (Glaeser, 2014).

Most of the studies focused on dams' qualitative evaluation except for de Faria et al. (2017), which examined the relationship between hydropower development and the socioeconomic conditions in Brazilian counties. Despite no significant effect on many social indicators, they found a negative impact on access to piped water and electricity, which could be explained by the growth in irregular housing. This result reinforces the findings of Duflo and Pande (2007) that state-level redistributive institutions failed to ensure the distribution of productive gains between the winners and losers of dam constructions.

I focused on the impact of the Belo Monte Hydroelectric Power Plant, the third largest in the world, constructed in the Pará State, Brazil. My analysis concerns the sudden change imposed by a large project on rapid and unplanned urbanization. I exploit the timing and distance of the dam's construction site to evaluate whether Belo Monte increased the violent crime level in its area of direct and indirect impact. By applying a difference-in-difference approach, I estimate the effect on the homicide rate during the construction of the Belo Monte dam (2011–2015) and after its conclusion (2016–2017). My regressions included all municipalities in the Pará state and control for municipalities fixed effects, time specific dummies, and covariates correlated with violent crime.

Using a panel dataset from 2007 to 2017, my main results show that Belo Monte dam construction led to a significant increase in the homicide rate in the municipalities included in the area of direct impact (ADI). The increase in violent crime persisted after its conclusion, indicating a negative inheritance of Belo Monte for the welfare of the region. The area of indirect impact (AII) does not present the same results as the ADI, which reinforces the theory that areas closer to the dam are the ones that bear the highest costs. My results prove that homicides are related to criminal activity and not motivated by other factors. Using illicit drug seizures, I show that the drug trafficking is one of the channels behind the rising homicide rate during Belo Monte construction. The results are robust to a set of checks that include serial correlation, the homicide rate time series trend, and the parallel trend assumption.

The evidence provided in this paper shows that the social cost of the Belo Monte dam includes a significant loss of young lives and a possible long-term violent conflict among drug trafficking organizations. These results align with the literature and show that policymakers and regulatory agencies should consider institutional capabilities when starting a large-scale project. The results are especially important for Brazil, which plans to build more than 40 hydroelectric dams in the Amazon region.

The remainder of the paper is structured as follows. [Section 2](#) provides the context of the Belo Monte dam and violence in Brazil. [Section 3](#) gives an overview of the related literature. [Section 4](#) presents the data used in the analysis. [Section 5](#) describes my empirical strategy. [Section 6](#) presents the results on the Belo Monte dam and violence. [Section 7](#) shows the robustness of the model. Finally, [Section 8](#) concludes the paper.

2. Context: Belo Monte dam

The project for the construction of the Belo Monte dam dates to 1975 with the Hydroelectric Inventory Studies for the Xingu river hydrographic basin. However, problems with other hydropower plants in the Amazon, were marked by lack of transparency, displacement of people, and flooding of indigenous territories weighted against the construction of Belo Monte. Also, the World Bank and other international organizations stopped funding large dams out of concern for environmental costs. Still, the country continues to grow, and after nationwide blackouts in 2001, the Belo Monte dam project was revised and entered into the Growth Acceleration Program of the Brazilian Federal Government (Moran, 2016).

Before the construction, much was discussed about the environmental costs of the Belo Monte dam (Fearnside, 1999). Nevertheless, less attention was given to the unpreparedness of all levels of government to deal with the negative social impacts of large-scale infrastructure projects. The dam construction attracted around 45,000 formal workers between 2011 and 2014, the equivalent of 46 percent of the Altamira population in 2010 (Miranda Neto, 2015). Although there is no official data about the number of migrants, it is possible to get a perspective of the size of the influx of people by looking at the number of formal employment in Figure 1, Panel A and B. The first figure shows the average level of formal jobs in the five municipalities (ADI) affected by the dam; the second one shows the level in Altamira, the city that concentrated the dam construction formal employment. The number of formal workers is only a fraction of the number of migrants, given that other informal employments follow the economic boom experienced in the region.

Urban land cover increased in Altamira between 2011 and 2016 as shown by Feng et al. (2017). I collected satellite images of Altamira to show the new neighborhoods created during the dam's construction to accommodate the displaced people that lived in the reservoir area. Besides the influx of workers, the region also had to incorporate *ribeirinhos* that used to live from natural resources along the Xingu River. Figure D3 and D4 in the [Supplementary Materials](#) show the city of Altamira in 2010 and 2014. One visible characteristic in Figure D4 (2014) is the urban sprawl segregating those local communities from the rest of the city.

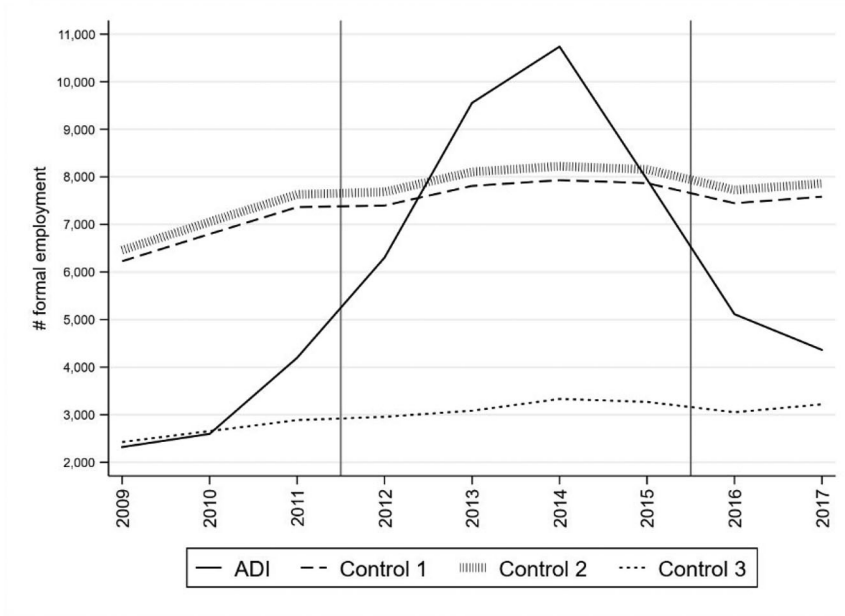
Improvements do not follow the sudden increase in population, given the lack of official figures for the number of inhabitants.¹ For instance, intergovernmental transfers depend on the official population size, which follows the official projection based on the 2010 census that completely disregards Belo Monte induced in-migration and resettlement in the region. The construction represented an overall loss of living conditions for the local population, with worse public health services (Grisotti, 2016), a lack of sanitation infrastructure (Gauthier & Moran, 2018), as well as housing and food inflation (Acevedo Marin & da Costa Oliveira, 2016; Bro, Moran, & Calvi, 2018; Calvi, Moran, Silva, & Batistella, 2020). Besides, in 2015, Altamira, one of the cities in the area of the direct impact of Belo Monte, was ranked as the most violent city in the country (Cerqueira et al., 2017).

The social costs of dam construction are often not well documented in the literature in the Global South. Still, they are described in the energy boomtown studies for the Global North (England & Albrecht, 1984; Freudenburg, 1981) and in the resource curse literature (van der Ploeg, 2011).

3. Related literature

While developed countries have stopped building dams because of increasing concerns about environmental and social impacts, developing countries are in the other direction investing in large hydropower plants to boost development. Hailed as a clean form of renewable energy, hydropower dams are being promoted as a pathway to social development and environmental protection (Lees, Peres, Fearnside, Schneider, & Zuanon, 2016; Wang, Tseng, & Zheng, 2015). In the case of Brazil, indirect benefits of building hydropower plants are promoted by the

Panel A. Number of formal employment in the area of direct impact (ADI)



Panel B. Number of formal employment in Altamira

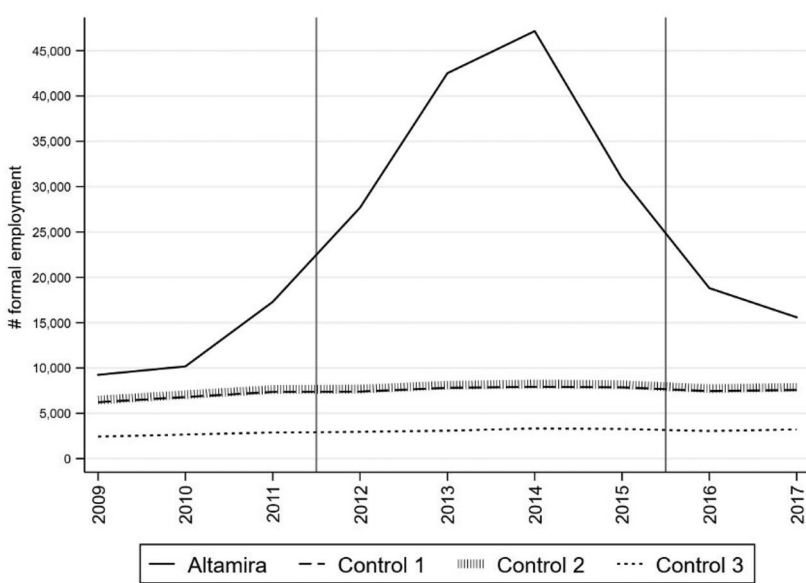


Figure 1. Number of formal employment in the ADI and Altamira (2007–2017).

Notes: The continuous lines show the yearly number of formal employment in the group of municipalities affected by Bale Monte (Area of direct impact - ADI) and in Altamira, the city that concentrated construction's formal employment. The other lines show the formal employment level in the other municipalities in the state of Pará, which serve as control groups for the analysis. The control groups vary to account for spill over and to exclude the metropolitan region of the state capital. In Panel A and B, Control 1 is formed by all municipalities except the ADI, Control 2 is formed by all municipalities except the ADI and the AII, and Control 3 is formed by all municipalities except the impacted areas (ADI + AII), the border municipalities and the metropolitan region.

policymakers and include improved infrastructure, such as roads, schools, hospitals, and economic compensation (de Faria et al., 2017; Eletrobras, 2009; Lees et al., 2016).

There is no unique conclusion on whether economic, social, and environmental benefits materialized (Ansar, Flyvbjerg, Budzier, & Lunn, 2014; de Faria et al., 2017; Fearnside, 2006, 2015; Jackson & Sleight, 2000; Tilt, Braun, & He, 2009). An important question that remains to be answered is how the costs and benefits of large dams are distributed between local communities and energy users (Bro et al., 2018; Duflo & Pande, 2007; Koch, 2002; Patil, Ghosh, & Kathuria, 2017; Soito & Freitas, 2011).

Another critical impact to be considered in hydropower developments in tropical regions, like the Amazon Basin, is the deforestation at the dam construction site and surrounding areas to open roads and create settlements. Deforestation is likely to impact the production of electric energy due to climate variability (Stickler et al., 2013), and environmental changes consequently produce social impacts as they result in loss of livelihood and poverty (Castro-Diaz, Lopez, & Moran, 2018; Soito & Freitas, 2011). For instance, as the riverine ecosystems decline in quality, so does the welfare of the local communities that depend on fisheries for food and income generation. Likewise, the study by Duflo and Pande (2007) about irrigation dams in India found that rural poverty increased in the district where the dam was built. Even with an increase in agricultural production and a decline in rural poverty in areas located downstream from the dam, the policy was considered cost-ineffective as it did not offset the negative impacts on the dam's district.

The social impacts of a hydropower plant incorporate the strain put on local infrastructure, health, and housing, caused by the influx of workers (Soito & Freitas, 2011) and the consequences such as sexually transmitted diseases (Grisotti, 2016) and drug use (von Sperling, 2012). Social cohesion is also negatively affected by the displacement of local communities that lived in the reservoir area as well as the encroachment by outsiders (Brown, Tullos, Tilt, Magee, & Wolf, 2009; Jackson & Sleight, 2000; von Sperling, 2012).

In the Belo Monte case, the hydropower plant acted as a catalyst for urban agglomeration. It attracted formal workers for the construction of the dam and people to be employed in indirect jobs created around the construction, coupled with displaced communities allocated within the closer cities. According to Glaeser and Sacerdote (1999) model, a city size increase is likely to increase crime rates. Moreover, under-developed regions² tend to have weaker institutions, making it even more challenging to deal with the impacts of large infrastructure projects (Glaeser, 2014).

Glaeser and Sacerdote (1999) investigate why cities have more crime. They decompose the connection between cities and crime into three categories. First, cities lower the costs of crime by reducing the probability of arrest; dense urban areas have a much larger number of suspects. Second, the cities' density produces higher returns to crime by creating proximity between potential wealthy victims and poor criminals. Third, urban areas attract crime-prone individuals. The same larger market size that makes cities appealing to firms also makes cities more appealing to drug dealers and thieves. Moreover, the authors argue that residents' preferences may be altered by city attributes, via social interactions and neighborhood effects, which make them more prone to crime (Glaeser, Sacerdote, & Scheinkman, 1996).

Social interactions are a key channel through which neighborhood crime is linked to individual criminal behavior (Damm & Dustmann, 2014). A critical remark in the literature is that the concentration of at-risk youth together in the same environment leads to higher crime levels. Therefore, neighborhood segregation increases crime by fostering social interactions among disadvantaged youth (Billings, Deming, & Ross, 2019) and negatively affects the future economic performance of the segregated groups (Cutler, Glaeser, & Vigdor, 2008).

Lastly, urbanization poses even more challenges for developing countries. Poorer countries tend to have weakly institutionalized contexts and fewer resources to deal with urban density externalities, such as congestion, contagious disease, and crime (Glaeser, 2014). The same weak

government poses a problem in controlling the spread of criminal organizations (Dell, Feigenberg, & Teshima, 2019).

4. Data

My unit of analysis is the administrative unit below a Brazilian state, a municipality. My sample includes all cities in the Pará state³ for the period between 2007 and 2017. The outcome variable is the number of homicides per 100,000 inhabitants, which indicates the incidence of violence (Fajnzylber, Lederman, & Loayza, 2002a, 2002b). All the mortality data employed in this paper come from death certificates and provide information on the cause, date, and place of occurrence.

I used variables related to the determinates of violent crime, which are available annually at the municipality level. The selected variables are meant to control employment level (number of formal employment, average earnings), socioeconomic conditions (number of cash-transfer program recipients, child mortality, GDP per capita), education (high school dropout rate), urbanization (suicides and mortality due to traffic accidents), and police effectiveness (police records of drug seizures).⁴

The treatment and control groups are defined based on the distance to the dam. The treatment group was determined in the environmental impact assessment (EIA) before the construction of the Belo Monte (Eletrobras, 2009). The EIA defined which cities would be directly impacted by the dam and the ones that would suffer an indirect influence. The studies pointed to two different areas for impact assessment, shown in Figure 2: Area of Direct Influence (ADI), with five municipalities⁵ and Area of Indirect Influence (AII), with seven cities.⁶ The two groups formed the target region of the Xingu Sustainable Regional Development Plan (PDRSX).

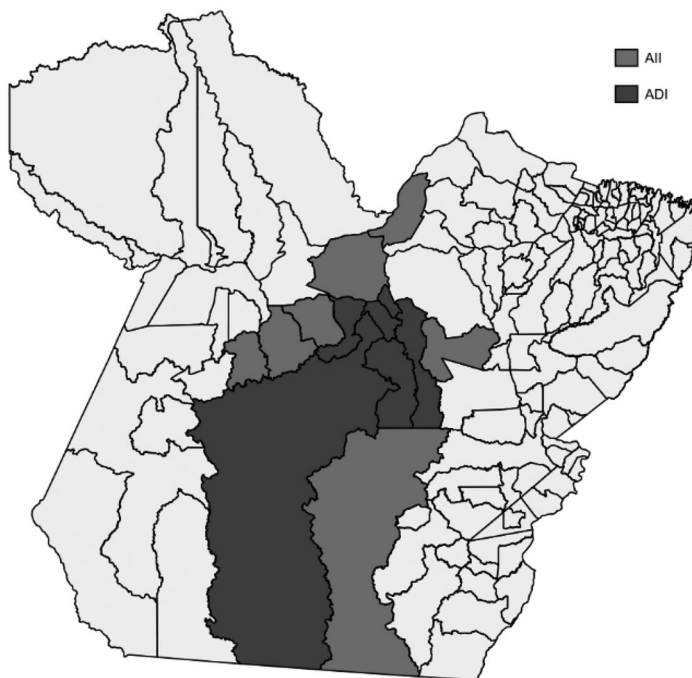


Figure 2. Municipalities impacted by Belo Monte dam – ADI and AII (Pará state).

Notes: The Environmental Impact Assessment (EIA) defined which cities would be impacted directly and indirectly by the dam. The area of direct influence (ADI) consists of 5 municipalities and the area of indirect influence (AII) has 7 municipalities.

I decided to use more than one control group to test whether there are spillovers of violence and the model's robustness when the state capital is excluded. The first control group includes all other municipalities except the treated ones. The second control group is conceived to avoid the spillover of violence since geographic proximity may lead to concerns that the control group might be contaminated. Surrounding municipalities are excluded from the second control group. For instance, the AII region will be excluded when the ADI is the treated area.

The different levels of characteristics that are determinants of violence, such as socioeconomic conditions and urbanization level, could affect the treatment effect. I propose a third control group that excludes the closest municipalities, the state's capital, and its metropolitan region. I use this control group as a robustness check for results and to have an equal control group to compare the ADI and AII. The control groups are shown in [Figure 3](#).

Panel A. Area of direct impact (ADI) of Belo Monte dam and control groups



Panel B. Area of indirect impact (AII) of Belo Monte dam and control groups

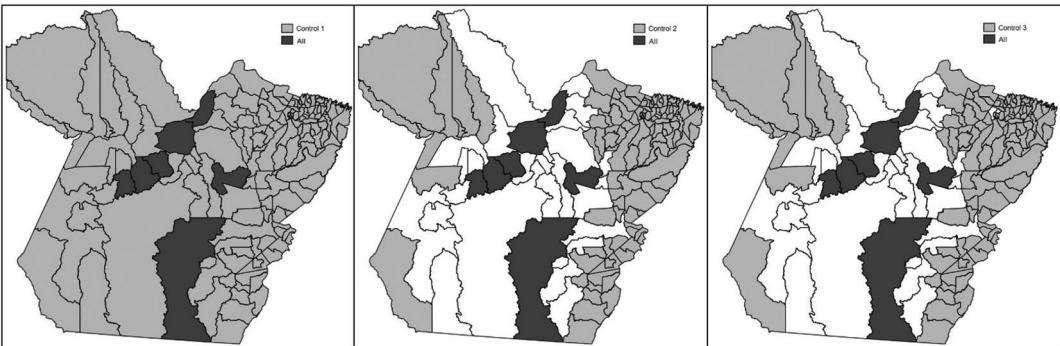


Figure 3. Control Groups for the ADI and AII (Pará state).

Notes: The control groups vary to account for spill over and to exclude the metropolitan region of the state capital. In Panel A, from left to right, Control 1 is formed by all municipalities except the ADI, Control 2 is formed by all municipalities except the ADI and the AII, and Control 3 is formed by all municipalities except the impacted area (ADI + AII), the border municipalities and the metropolitan region. In Panel B, from left to right, Control 1 is formed by all municipalities except the AII, Control 2 is formed by all municipalities except the AII, ADI and border municipalities, and Control 3 is formed by all municipalities except the impacted areas (ADI + AII), the border municipalities and the metropolitan region. The Control 3 includes the same municipalities for the ADI and AII.

5. Empirical strategy

Following a difference-in-difference (DiD) approach, I explore the timing of Belo Monte dam construction and the distance from the construction site to identify the causal effect of unplanned urbanization on the violent crime level. Given the previous discussion, I focus on two different moments: the start of the construction (June 2011) and the subsequent rapid urbanization of the region, and the end of construction followed by a massive layoff (November 2015).

The model to be estimated is the following difference-in-difference regression:

$$\begin{aligned} Homicide_{it} = & \alpha + \beta_1 \cdot (D_{2011 \leq t \leq 2015} \times treated_i) \\ & + \beta_2 \cdot (D_{2016 \leq t \leq 2017} \times treated_i) \\ & + z'_{it}\delta + \theta_i + \gamma_t + \varepsilon_{it} \end{aligned}$$

where Homicide is the homicide rate for municipality i in year t ; $D_{2011 \leq t \leq 2015}$ is a dummy variable equal to 1 for the years between 2011 and 2015; $D_{2016 \leq t \leq 2017}$ is a dummy variable equal to 1 between 2016 and 2017; $treated$ is a dummy variable equal to 1 if the municipality is in the impacted regions; z_{it} is a set of municipality-level control variables; θ_i is a municipality fixed effect; γ_t is a time fixed effect; ε_{it} is a random term; and α , β_1 , β_2 , and δ are parameters to be estimated.

One concern is that the dam's construction also affects the municipality's characteristics which are determinants of violent crime. Therefore, the control variables are themselves outcome variables. According to Angrist and Pischke (2009), controlling for outcome variables can be misguided, giving origin to the potential problem of 'bad controls'. The control variables should be measured before the variable of interest is determined. Hence, I decided to use as control variables the interactions between pre-treatment values (2010) with time dummies instead of directly controlling for their contemporaneous values. This procedure would control for pre-treatment municipality characteristics that are time-variant and potentially correlated with violent crime and other covariates.

The set of control variables that are interacted with time dummies are number of formal employment, average earnings (ln), number of cash-transfer program recipients (*Bolsa Família*), child (before age five) mortality rate, GDP per capita (ln), high school dropout rate, drug trafficking crime records, suicides and mortality due to traffic accidents. Also, it is critical to include an interaction between pre-treatment values of homicide rate (2010) and time dummies to account for criminal inertial. The model allows different dynamics of violence conditional on municipality characteristics before the implementation of Belo Monte dam.

To address concerns related to the trend in the homicide time series, I run a specification that allows the treatment to affect both the level and the trend of the outcome variable to assess whether the trend could be affecting the violence level. Additionally, I tested if the results were driven by a spurious regression and use the lagged values ($t-1$) of the homicide rate as control variable instead of the interaction mentioned above. I also test whether the parallel trend assumption holds by creating a pre-intervention placebo variable for years before Belo Monte, followed by a graph plotting a flexible DiD regression with leads and lags.

Additionally, since the size of the population is directly related to the variance of the homicide rate, all regressions are weighted by population size. The standard errors are clustered at the municipality level to avoid underestimation due to autocorrelation in the residuals (Bertrand, Duflo, & Mullainathan, 2004).

6. Results

For descriptive purposes, in [Figure 4](#), Panel A and B, plot the yearly homicide rate for the municipalities in the state of Para between 2007 and 2017, disaggregated by treated areas and their respective controls. The panels indicate the two shocks: first, in June 2011, when the construction of Belo Monte started, and second, between November 2015 and January 2016, when the construction ended, and a massive layoff took place ([Oliveira, 2017](#)).

In [Figure 4](#), panel A, the ADI has a similar trend to the control groups in 2009 and 2010. Although the homicide rate in the control groups continues to increase over time, the ADI experienced explosive growth in violence. In the AII, panel B, the parallel trend assumption in homicide rate before 2011 does not seem to hold. For this reason, the results concerning AII are included in Section E in the [Supplementary materials](#).

6.1. Main results

[Table 1](#) presents the ADI's main results. The column titles indicate which control group is used. For each control group, the results are shown without and with interactions of time dummies with pre-intervention values of the set of covariates plus homicide rate.⁷

All columns show a significant effect on violence in both shocks, the start (Treated ADI $\times$ Post 2011) and the end of the construction (Treated ADI $\times$ Post 2016). On average, the five municipalities included in the ADI experienced an increase in violent crime during the Belo Monte dam construction, followed by a more intense increase after its conclusion.

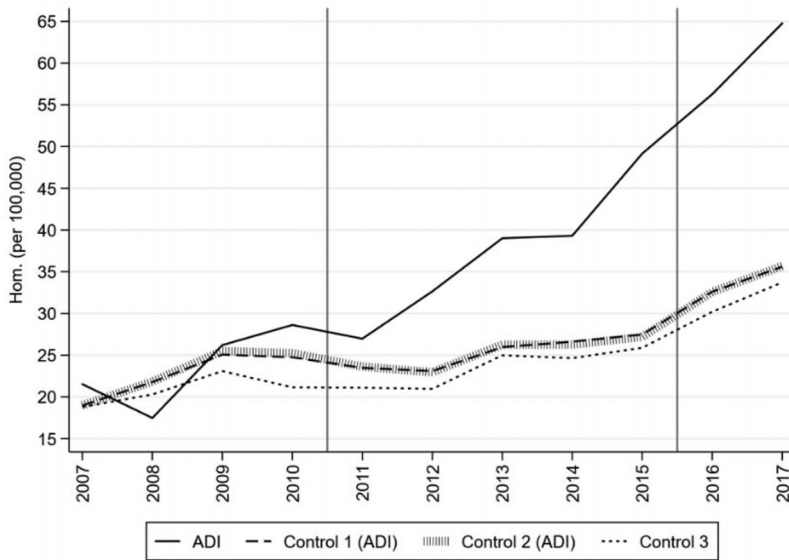
Control group 2 is a robustness check for violence spillovers between treatment and control municipalities. If there is such violence spillover, the coefficients on columns 3 and 4 should be higher than the ones in columns 1 and 2, respectively. Once surrounding municipalities are excluded from the control group, the coefficients should capture the effect of the Belo Monte dam on violent crimes more accurately. The small difference between columns 2 and 4 suggests the presence of some spillovers and an underestimation when using control group 1. Control group 3 is a robustness test for the previous coefficients, as it does not include a larger number of surrounding municipalities ([Figure 3](#), Panel A) and the metropolitan region. The results in columns 5 and 6 continue to show consistent results of the impact of the Belo Monte dam on the homicide rate in the ADI.

It is also worth noting that in columns 2, 4 and 6 when I include the set of interacted controls, there is a change in point estimates in both shocks. Therefore, a difference between treated municipalities and control groups is driven by differential pre-intervention characteristics. To analyze what characteristic is affecting the behavior of homicide rates, I separated the set of covariates into groups of variables relating to the determinates of crime and estimated different specifications accounting for each of them separately.

[Table 2](#) shows the results using control group 2, the main control group. Columns 1 and 9 are the same as columns 3 and 4 of [Table 1](#). The set of covariates is separated and included in different columns: column 2 contains the education variable, column 3 includes urbanization variables, column 4 holds constant socioeconomic conditions, column 5 includes employment variables, column 6 includes drug trafficking variable, and column 7 contains the homicide rate in 2010 interacted year dummies. Column 8 shows the coefficients when all covariates are included, excluding the homicide rate. Overall, the only characteristic that affects point estimates is a different initial level of urbanization across treated and control municipalities.

For the quantitative implication, considering the coefficients in column 4 of [Table 1](#), the coefficients can be read as changes in homicide rate per 100,000 inhabitants during the construction of Belo Monte dam and after its conclusion, indicating average increases of, respectively, 38.11 per 100,000 inhabitants between 2011 and 2015, 58.14 after 2016. Comparing these with pre-Belo Monte average homicide rate in the ADI, the construction period raised the violent crime level by 163 percent and the two years following the massive layoff had an average increase of

Panel A. Homicide rate in the area of direct impact (ADI)



Panel B. Homicide rate in the area of indirect impact (AII)

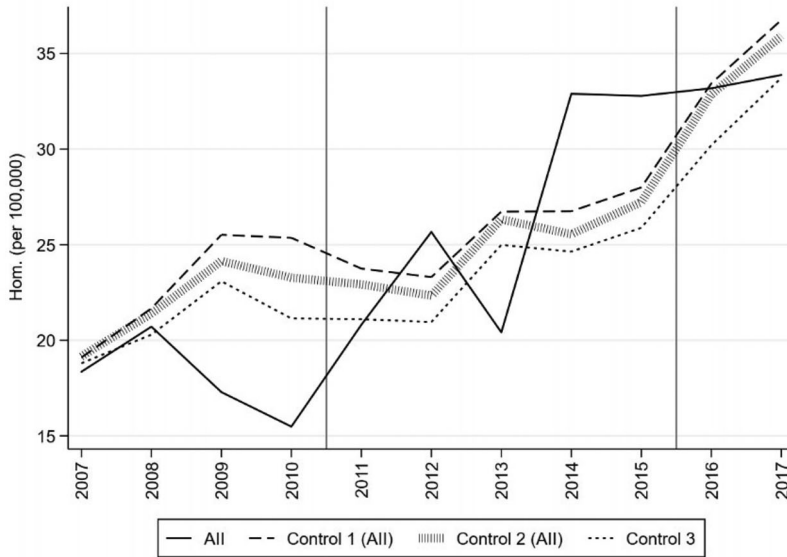


Figure 4. Homicide rate in the impacted areas of Belo Monte dam (2007–2017).

Notes: The control groups vary to account for spill over and to exclude the metropolitan region of the state capital. In Panel A (ADI), Control 1 is formed by all municipalities except the ADI, Control 2 is formed by all municipalities except the ADI and the AII, and Control 3 is formed by all municipalities except the impacted areas (ADI + AII), the border municipalities and the metropolitan region. In Panel B (AII), Control 1 is formed by all municipalities except the AII, Control 2 is formed by all municipalities except the AII, ADI and border municipalities, and Control 3 is formed by all municipalities except the impacted areas (ADI + AII), border municipalities and the metropolitan region. First shock is in June 2011 when the construction started (Post 2011, 2011–2015) the second shock is between November 2015 and January 2016, when it happened a massive layoff (Post 2016, 2016–2017).

Table 1. Main results for the ADI – Belo Monte dam and Homicides, 2007–2017, difference-in-difference

Variables	(1) Control 1	(2) Control 1	(3) Control 2	(4) Control 2	(5) Control 3	(6) Control 3
Treated ADI × Post 2011	26.69** (10.51)	37.28*** (10.67)	26.86** (10.54)	38.11*** (10.68)	26.70** (10.46)	34.61*** (11.49)
Treated ADI × Post 2016	44.02*** (12.26)	56.70*** (11.21)	44.06*** (12.30)	58.14*** (11.18)	47.80*** (12.23)	57.31*** (13.85)
Constant	29.13*** (3.376)	29.16*** (1.483)	29.37*** (3.474)	29.40*** (1.516)	23.85*** (1.960)	23.83*** (1.628)
Municipality FE	X	X	X	X	X	X
Year FE	X	X	X	X	X	X
Set of covariates		X		X		X
Observations	1,571	1,571	1,494	1,494	1,219	1,219
R-squared	0.806	0.875	0.806	0.876	0.719	0.783

Notes: Robust standard errors are in parentheses (clustering at municipality); *** $p < 0.01$, ** $p < 0.05$. Dependent variable is the homicide rate (per 100,000 inhabitants). All regressions are weighted by population. Treatment variables are dummies = 1 if the municipality is part of the area of direct influence of Belo Monte dam interacted with post intervention dummies: post 2011, if between 2011–2015 and post 2016, after 2016. Set of covariates represents the interactions of year dummies with pre-intervention (2010) values of the following municipality characteristics that are determinants of violent crime: number of formal employment, average earnings (ln), number of cash-transfer program recipients, child mortality, GDP per capita (ln), high school dropout rate, mortality by traffic accidents, suicides, drug trafficking, and homicide rate. The columns titles indicate which control group is used. The control groups vary to account for spill over and to exclude the metropolitan region of the state capital: Control 1 is formed by all municipalities except the ADI, Control 2 is formed by all municipalities except the ADI and the AII, and Control 3 is formed by all municipalities except the impacted areas (ADI + AII), the border municipalities and the metropolitan region.

248 percent. These are significant increases considering that the control municipalities also experienced a rise in violence levels, as shown in the descriptive statistics in [Table B1](#) in the [supplementary materials](#).

To support the theory of an increase in violent crime levels in the ADI of the Belo Monte dam, I break down the dependent variable by victims' characteristics. Restricting the analysis to specific demographic characteristics helps shed light on how violence affects different groups. [Table 3](#) presents the results and shows that the violence increase affected mainly men, especially young males aged 15–29. Additionally, it is possible to observe the progression of homicides by firearm and homicides that occurred in the public way (excluding the possibility of domestic violence). [Table 3](#) shows the average homicide rate level before Belo Monte dam (2007–2010) and the proportional effect of each shock.

6.2. Discussion

The results in [Tables 1](#) and [3](#) show that the Belo Monte dam affected the violent crime level in closer municipalities. The coefficients seem to follow the literature on the economics of crime. The first increase in violence (2011–2015) is driven by urbanization and its negative externalities (e.g. drug trafficking). The second increase (2015–2016) incorporates the effect of higher unemployment, criminal inertia, and social interactions. The results undoubtedly prove that violence is not related to domestic violence or family disputes but criminal activity in the region.⁸

Rapid urbanization leads to negative externalities. [Figure 1](#), in the context section, shows the population increase in the ADI and Altamira.⁹ Besides, the analysis of the data used as a proxy for urbanization (suicides and mortality due to traffic accidents) presents a significant increase

Table 2. Covariates – Belo Monte dam and Homicides, 2007–2017, difference-in-difference

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
		Education	Urbanization	Socioeconomic	Employment	Drug traff.	Homicide		Main result
Treated ADI × Post 2011	26.86** (10.54)	26.83** (10.61)	33.94*** (12.42)	27.14** (11.37)	26.83** (10.36)	26.62** (10.67)	26.88** (10.82)	38.67*** (10.89)	38.11*** (10.68)
Treated ADI × Post 2016	44.06*** (12.30)	43.95*** (12.35)	53.34*** (14.59)	44.99*** (12.92)	47.04*** (11.33)	46.66*** (12.31)	44.09*** (12.67)	59.30*** (11.76)	58.14*** (11.18)
Constant	29.37*** (3.474)	29.37*** (3.430)	29.37*** (3.490)	29.37*** (3.514)	29.36*** (3.479)	29.37*** (3.491)	29.38*** (2.317)	29.40*** (1.799)	29.40*** (1.516)
Municipality FE	X	X	X	X	X	X	X	X	X
Year FE	X	X	X	X	X	X	X	X	X
Education (2010)		X						X	X
Urbanization (2010)			X					X	X
Socioeconomic (2010)				X				X	X
Employment (2010)					X			X	X
Drug trafficking (2010)								X	X
Homicide rate (2010)						X		X	X
Observations	1,494	1,494	1,494	1,494	1,494	1,494	X	1,494	1,494
R-squared	0.806	0.807	0.818	0.816	0.814	0.811	0.832	0.861	0.876

Notes: Robust standard errors are in parentheses (clustering at municipality); *** $p < 0.01$, ** $p < 0.05$. Dependent variable is the homicide rate (per 100,000 inhabitants). All regressions are weighted by population. Treatment variables are dummies = 1 if the municipality is part of the area of direct influence of Belo Monte dam interacted with post intervention dummies: post 2011, if between 2011–2015 and post 2016, after 2016. Set of covariates represents the interactions of year dummies with pre-intervention (2010) values of the following municipality characteristics that are determinants of violent crime: number of formal employment, average earnings (ln), number of cash-transfer program recipients, child mortality, GDP per capita (ln), high school dropout rate, mortality by traffic accidents, suicides, and homicide rate. The set of covariates is separated by characteristics determinants of violent crime: Education (high school dropout rate), Urbanization (suicides and mortality due to traffic accidents), Socioeconomic Conditions (number of cash-transfer program recipients, child mortality, GDP per capita), Employment level (number of formal employment, average earnings), drug trafficking, plus homicide rate. All control variables are pre-intervention (2010) values interacted with year dummies. All columns use the control group 2 that is formed by all municipalities except the ADI and the AII to account for violence spillover.

Table 3. Victims characterization – Belo Monte dam and Homicides, 2007–2017, difference-in-difference

Variables	Homicide		Female	Male Homicide		Homicide	Male Homicide (ages 15–29)		
	All (1)	Homicide All (2)	All (3)	By firearm (4)	Public way (5)	All (6)	All (7)	By firearm (8)	Public way (9)
Treated ADI × Post 2011	38.11*** (10.68)	1.017 (2.091)	77.67*** (21.32)	48.43*** (13.56)	37.61*** (9.785)	139.9*** (23.36)	239.4*** (48.19)	147.1*** (43.87)	198.8*** (24.41)
Treated ADI × Post 2016	58.14*** (11.18)	7.019 (4.981)	118.4*** (18.85)	77.76*** (16.62)	50.92*** (9.820)	187.7*** (27.81)	309.9*** (50.41)	254.2*** (60.57)	270.1*** (31.65)
Constant	29.40*** (1.516)	3.994*** (0.330)	59.92*** (3.218)	38.44*** (2.211)	29.36*** (1.553)	143.3*** (5.062)	224.8*** (8.050)	122.4*** (6.205)	178.0*** (7.416)
Avg. level (ADI) 2007–2010	23.45	5.55	50.32	27.30	19.58	109.63	169.76	90.50	187.99
Proportional effect during the construction (2011–2015)	163%	0	154%	177%	192%	128%	141%	163%	106%
Proportional effect after the construction (2016–2017)	248%	0	235%	285%	260%	171%	183%	281%	144%
Municipality FE	X	X	X	X	X	X	X	X	X
Year FE	X	X	X	X	X	X	X	X	X
Set of covariates	X	X	X	X	X	X	X	X	X
Observations	1,494	1,494	1,494	1,494	1,494	1,494	1,494	1,494	1,494
R-squared	0.876	0.538	0.877	0.897	0.839	0.888	0.890	0.913	0.805

Notes: Robust standard errors are in parentheses (clustering at municipality); *** $p < 0.01$. Dependent variable is the homicide rate (per 100,000 inhabitants) by demographic group. The denominator is the total population of each subgroup. All regressions are weighted by population. Treatment variables are dummies = 1 if the municipality is part of the area of direct influence of Belo Monte dam interacted with post intervention dummies: post 2011, if between 2011–2015 and post 2016, after 2016. Set of covariates represents the interactions of year dummies with pre-intervention (2010) values of the following municipality characteristics that are determinants of violent crime: number of formal employment, average earnings (ln), number of cash-transfer program recipients, child mortality, GDP per capita (ln), high school dropout rate, drug trafficking, mortality by traffic accidents, suicides, and homicide rate. All columns use control group 2 which is formed by all municipalities except the ADI and the All to account for violence spillover.

in urbanization levels during and after the dam construction; results are shown in Table D1 (supplementary materials). Communities living in the reservoir (*ribeirinhos*) were relocated around Altamira¹⁰ and faced new challenges. First, they are far from the river where they used to fish and grow food, making them urban poor and dependent on social aid (Brum, 2018). Second, the new housing arrangement did not respect their previous way of living by putting together opposing tribes and communities (Cerqueira et al., 2017). Social Capital is found to be reduced in both resettled and host populations, which nurtures violence and conflict within a community, according to the literature (Mayer, Lopez, Cavallini Johansen, & Moran, 2022; Mayer, Lopez, Leturcq, & Moran, 2022).

The increase in population and income in the region is followed by the rise in illicit drug seizures, used as a proxy for the level of drug trafficking. Figure 5, Panel A and B show the annual seizure rate per 100,000 inhabitants in the ADI and Altamira. It is possible to observe explosive growth in the cities affected by the Belo Monte compared to the rest of the Pará state. To investigate the role of drug trafficking in the ADI and Altamira, I run a DiD regression and use the variable illicit drug seizures as outcome. The results are shown in Table 4. Drug trafficking in the ADI and Altamira increased during the construction period. Brazil has many criminal organizations that fight for market share, creating more violence in the cities where they are present (Biderman, De Mello, De Lima, & Schneider, 2019). That seems to be the case for the ADI once the dam construction raised the income level (Ferreira, 2019; Phillips, 2019).

The years following the end of construction continued to have high levels of violent crime, which may be explained by a worse job market and unemployment. From Figure 1, it is possible to see a decrease in job opportunities. Even though some of the workers are migrants that could have gone back to their hometowns, Belo Monte construction company information says that at least 3,332 of the workers dismissed at the end of 2015 were residents of Altamira (Oliveira, 2017). This result is in line with the recent evidence that the probability of committing crimes increases on average by 23 percent for workers displaced by mass layoffs, with a 55–58 percent increase in drug-related crimes and a 32 percent increase in homicides (Britto, Pinotti, & Sampaio, 2022).

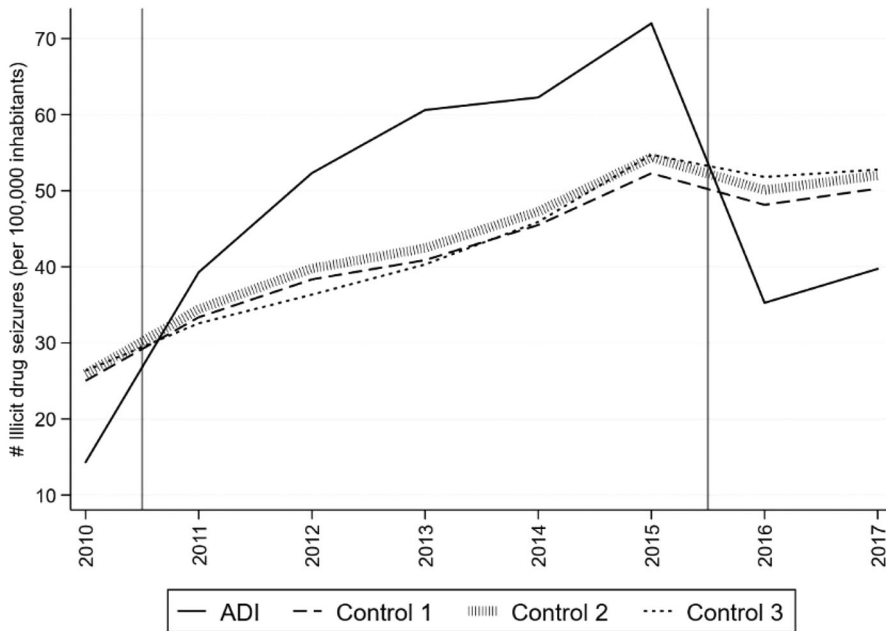
Also, the previous level of violence (during the construction period) affects the violence in the following years. This phenomenon is called by Fajnzylber et al. (2002b) of criminal inertia, where the past incidence of crime in society affects the individual's decision to commit a crime by reducing the costs of carrying out criminal activities, lowering the perceived probability of arrest, and weakening civic moral values. Moreover, vulnerable young people were segregated in the new neighborhoods. According to an official government report, the effect of social interactions among these individuals is likely to have enhanced their involvement in crime (Cerqueira et al., 2019). This is aligned with the literature in which neighborhood segregation increases crime by fostering social interactions among disadvantaged youth (Billings et al., 2019).

7. Robustness check

To assess how much the trend is responsible for the violence increase, I allow the treatments to affect both the level and trend of the dependent variable by interacting each treatment with a linear time trend equal to zero in the first year of treatment. Column 1 of Table 5 suggests that during the construction of Belo Monte dam in the first period, there was a higher increase in the homicide rate than in the trend. In the second period, after the end of the construction, there was a significant increase compared to the previous level (2011–2015), and the trend is not significant. However, the homicide rate time series has only two years after the construction (2016–2017), and the coefficient of the second trend may not be accurately estimated.

Due to this time limitation in the dataset, I further document whether the trend and, consequently, the serial correlation may bias the results. To avoid a spurious relationship,

Panel A. Drug trafficking rate in the area of direct impact (ADI)



Panel B. Drug trafficking rate in Altamira

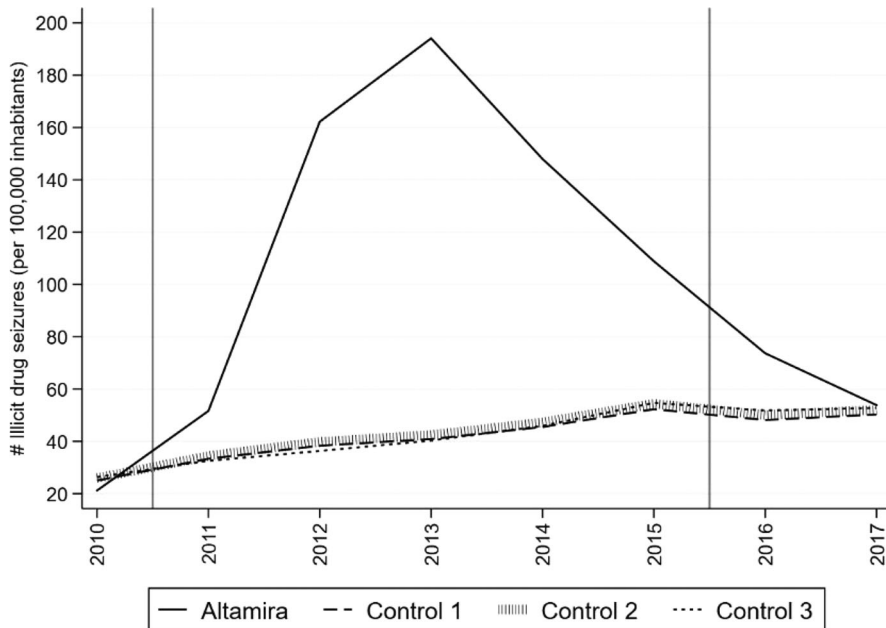


Figure 5. Drug trafficking rate in the ADI and Altamira (2010–2017).

Notes: The control groups vary to account for spill over and to exclude the metropolitan region of the state capital. In Panel A and B, Control 1 is formed by all municipalities except the ADI, Control 2 is formed by all municipalities except the ADI and the AII, and Control 3 is formed by all municipalities except the impacted areas (ADI + AII), the border municipalities and the metropolitan region.

Table 4. Drug trafficking in the ADI and Altamira, 2010–2017, difference-in-difference

Variables	(1) Control 2	(2) Control 2
Altamira × Post 2011		89.26*** (5.266)
Altamira × Post 2016		21.40*** (7.684)
Treated ADI × Post 2011	58.50** (23.87)	
Treated ADI × Post 2016	8.946	
Constant	50.19*** (1.851)	50.25*** (1.768)
Municipality FE	X	X
Year FE	X	X
Set of Covariates	X	X
Observations	1,088	1,056
R-squared	0.735	0.738

Notes: Robust standard errors are in parentheses (clustering at municipality); *** $p < 0.01$, ** $p < 0.05$. Dependent variable is illicit drug seizures (per 100,000 inhabitants). All regressions are weighted by population. Treatment variables are dummies = 1 if the municipality is part of the area of direct influence of Belo Monte dam or Altamira interacted with post intervention dummies: post 2011, if between 2011–2015 and post 2016, after 2016. Set of covariates represents the interactions of year dummies with pre-intervention (2010) values of the following municipality characteristics that are determinants of violent crime: number of formal employment, average earnings (ln), number of cash-transfer program recipients, child mortality, GDP per capita (ln), high school dropout rate, mortality by traffic accidents, suicides, and homicide rate. Both columns use the control group 2 that is formed by all municipalities except the ADI and the AII to account for violence spill over.

I re-estimate the same specification from Table 1, column 4, but now using the set of covariates with the lagged values of homicide rate (t-1) instead of its pre-intervention value (2010) interacted with year dummies. The results are presented in column 2 of Table 5. The results are still statistically significant, but the point estimate has changed since it accounts for the homicide time series trend. Nevertheless, the effect of Belo Monte dam on the violence level is robust and not driven by a spurious correlation.

As a final test of the model and the DiD approach, I test the parallel trend assumption to rule out the possibility that the treatment variables capture distinct pre-existing dynamics of violence in the ADI. If this were the case, a placebo indicating pre-Belo Monte years would detect whether homicides in the ADI were already increasing a couple of years before the construction started. The results of this exercise are in column 3 of Table 5. I included a dummy for 2009–2010 interacted with ADI municipalities, Pre-Belo Monte variable. The coefficient for the pre-intervention period is very small and not statistically significant. Therefore, there is no evidence that the treatment effects capture the differential dynamic behaviour of homicide rates before the Belo Monte dam.

Continuing the tests on the parallel trend assumption, I estimated flexible DiD regression with leads and lags, where the treated municipalities are interacted with year dummies. The interaction terms are meant to measure the deviation in trends in each period and detect the specific timing of the differential behavior of homicides across treated and control municipalities. The ten coefficients estimated, with their respective confidence intervals, are plotted in Figure 6. The homicide rate level is not statistically different from zero in the years before Belo Monte. The difference in homicide rates across treated and control municipalities are statistically significant from 2012, one year after construction started. There was a small decrease in point estimate in 2016, the first year after the massive layoff that marked the end of the construction, followed by a new increase in 2017.

Table 5. Testing trend and parallel trend assumption – Belo Monte dam and Homicides, 2007–2017, difference-in-difference

Variables	Linear trend (1)	Lagged values (2)	Pre-Intervention (3)
Treated ADI × Post 2011	21.68*** (8.269)	31.57*** (10.20)	38.36*** (9.607)
Treated ADI × trend (2011–2015)	8.214*** (1.923)		
Treated ADI × Post 2016	45.46*** (11.32)	45.60*** (11.23)	58.38*** (10.62)
Treated ADI × trend (2016–2017)	25.35 (25.15)		
Pre-Belo Monte (2009–2010)			0.489 (6.053)
Constant	29.40*** (1.518)	37.16*** (2.555)	29.40*** (1.517)
Municipality FE	X	X	X
Year FE	X	X	X
Set of Covariates	X	X	X
Observations	1,494	1,357	1,494
R-squared	0.878	0.882	0.876

Notes: Robust standard errors are in parentheses (clustering at municipality); *** $p < 0.01$. Dependent variable is the homicide rate (per 100,000 inhabitants). All regressions are weighted by population. Treatment variables are dummies = 1 if the municipality is part of the area of direct influence of Belo Monte dam interacted with post intervention dummies: post 2011, if between 2011–2015 and post 2016, after 2016. Set of covariates represents the interactions of year dummies with pre-intervention (2010) values of the following municipality characteristics that are determinants of violent crime: number of formal employment, average earnings (ln), number of cash-transfer program recipients, child mortality, GDP per capita (ln), high school dropout rate, drug trafficking, mortality by traffic accidents, suicides, and homicide rate (homicide rate pre-intervention values interacted with year dummies in columns 1 and 3. Column 2 includes the lagged values (t-1) of homicide rate). Treated ADI × trend is an interaction between treatment municipalities, post intervention and linear time trends equal to zero in the first year of treatment. Pre-Belo Monte placebo is a dummy for 2009–2010 interacted with treated ADI municipalities. Column 2 includes the same set of covariates, but the homicide rate included as control variable is the lagged value (t-1). All columns use the control group 2 that is formed by all municipalities except the ADI and the AII to account for violence spill over.

I further tested the robustness of the model and placed the results in the [supplementary materials](#) (Table D5 and Table D6).

8. Conclusion

This paper sheds light on how development policies could have unexpected consequences. The results are in line with previous literature and show that urbanization is followed by negative externalities. Without strong institutions that can cope with those externalities, urbanization can bring chaos and welfare loss. My results are relevant to other developing countries where violence is a systemic issue and plan to implement large-scale projects that have the potential of creating boomtowns.

Adding to the previous finding of de Faria et al. (2017) that hydropower plants in Brazil generate only short-term economic development, this paper shows that the economic boom also represents, at least, a short-term increase in violent crime. Since violence and crime hinder growth (da Mata, Deichmann, Henderson, Lall, & Wang, 2007), Belo Monte dam and the Xingu Sustainable Regional Development Plan (PDRSX) failed to deliver local development and may have caused a systemic problem in the region.

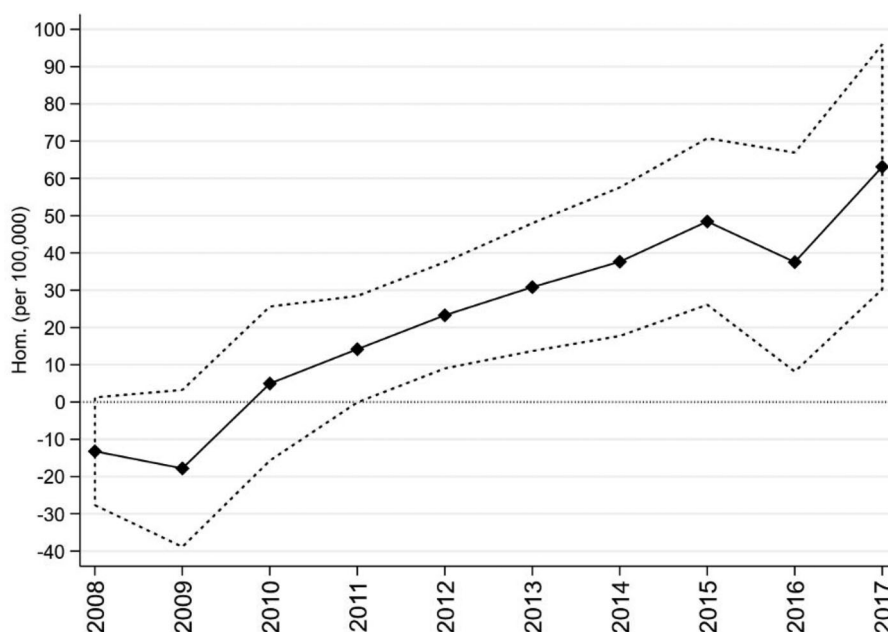


Figure 6. Timing of the effect, ADI (2008–2017).

Note: 95 percent confidence interval.

The northern states of Brazil have been known to be routes for international drug trafficking, and the ADI is now included in those routes (Cerqueira et al., 2019; Ferreira, 2019). Rapid urbanization in the ADI and Altamira raised drug trafficking, which challenges long-term development. Results presented showed a decrease in drug seizures around Belo Monte region. However, drug gangs exist in Altamira and continue fighting for market share (Phillips, 2019). The mass layoff and the presence of drug gangs in the area are probably associated with the long-lasting violence following the end of the construction. Future research should look into the long-term social effects of large-scale projects, especially those in underdeveloped regions.

Data availability imposed obstacles for this study since proxy variables used as covariates have their limitations. The lack of data was even one of the issues that contributed to the negative consequences of Belo Monte. Policymakers could have anticipated the need for timely data to effectively implement and monitor the PDRSX. Without an extra budget, the cities could not cope with the rapid increase in population size. Although the PDRSX plan included the obligation for Belo Monte owner company to invest in public infrastructure (e.g. schools and hospitals), these were not delivered in time. Another issue was the lack of resources dedicated to improving public security (Oliveira, 2017). New projects should thoroughly consider the social component of sustainable development and develop a comprehensive plan to avoid predictable externalities, as in the Belo Monte case.

Belo Monte dam was set to be an example that the exploitation of Amazon resources could be combined with local sustainable development. The results of this paper raise doubts about the development plan implemented and produce evidence that could help inform decisions on the other infrastructure projects planned for the Brazilian Amazon. The increase in violence represents a loss of human capital, even more so since young people are the most affected. Violence imposes significant social costs and may jeopardize future growth in the Amazon. Without proper institutions and resources, development projects can end up intensifying social losses.

Notes

1. According to the 2010 population census Altamira had about 99,075 inhabitants and official projection for 2015 was 108,382 inhabitants. However, the Municipality of Altamira estimated 148,224 inhabitants already in 2012 based on access to medical and hospital services (Neto & Herrera, 2016).
2. For an overview of socioeconomic indicators in Pará and the rest of Brazil see Figure C1 (Supplementary Materials).
3. I use only the state where the Belo Monte dam is located (Pará) since the police (deterrence effect) is organized at the state level. Furthermore, municipalities are more homogeneous within one state.
4. Dataset details are in Table A1 (Supplementary materials).
5. Altamira, Anapu, Brasil Novo, Senador José Porfírio and Vitória do Xingu.
6. Gurupá, Medicilândia, Pacajá, Placas, Porto de Moz, São Félix do Xingu and Uruará.
7. For robustness purposes, I estimate the model using the contemporaneous values of the set of covariates and the results are in Table D2 (Supplementary materials).
8. Other types of violence (domestic violence and rape) in the region are reported in Figures D1 and D2 (Supplementary materials).
9. I also run a DiD regression for Altamira to estimate the impact of the Belo Monte dam on homicide rate and the results are in Table D4 (Supplementary materials).
10. See Figures D3 and D4 in Supplementary materials.

Acknowledgments

All data and code used to estimate the results in the paper are openly available in figshare at <https://doi.org/10.6084/m9.figshare.22012562.v1>

Disclosure statement

No potential conflict of interest was reported by the author(s).

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